

SOP-04: Formaldehyde Solution 37% / Formalin (PHS)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Formaldehyde solution 37% (formalin, F75F-1GAL)	50-00-0
Working 10% neutral buffered formalin / 4% formaldehyde fixative prepared from the above	50-00-0

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Health hazard (carcinogen, sensitizer, STOT); Skull and crossbones (acute toxicity Cat 3); Corrosion (skin corrosion Cat 1B, eye damage Cat 1); Flame (flammable liquid Cat 3)

- PARTICULARLY HAZARDOUS SUBSTANCE. Formaldehyde is a GHS Carcinogenicity Category 1A substance (known human carcinogen; IARC Group 1; NTP Known to be a Human Carcinogen), a germ cell mutagen Category 2, a respiratory and skin sensitizer, and is acutely toxic by all three routes. Sections 9-11 of this SOP are mandatory.
- Formaldehyde is additionally regulated under a substance-specific standard: WISHA WAC 296-856 (Formaldehyde) / OSHA 29 CFR 1910.1048. The 8-hour TWA PEL is 0.75 ppm, the 15-minute STEL is 2 ppm, and the Action Level is 0.5 ppm. Exposure monitoring, medical surveillance, hazard signage and specific training are required if exposures may reach the Action Level or STEL. Contact EH&S before beginning any new formaldehyde operation so that an exposure assessment can be performed.

- Stock chemicals: 1 gallon of 37% formaldehyde solution (stabilized with methanol). Intermediates and working solutions: 10% neutral buffered formalin (approximately 3.7% formaldehyde) and 4% paraformaldehyde-equivalent fixative prepared by dilution in phosphate buffer or filtered seawater, used to fix bivalve and finfish tissue. Wastes: spent fixative containing tissue, formaldehyde rinses, and formaldehyde-contaminated solid debris.
- Routes of exposure: inhalation of vapor (primary and most significant), skin contact and absorption, eye contact, ingestion.
- How exposure might occur: opening the gallon container, pouring and diluting the 37% stock, filling and topping off specimen jars, decanting spent fixative from tissue jars, handling fixed tissue, and opening long-closed specimen containers in which vapor has equilibrated.
- Target organs: upper respiratory tract (nose, nasopharynx), eyes, skin, and the immune system. Formaldehyde causes nasopharyngeal cancer and is associated with myeloid leukemia.
- Signs and symptoms of exposure: burning of the eyes, nose and throat; tearing; cough; wheezing; chest tightness; headache and nausea. Skin contact causes whitening, hardening and cracking, allergic contact dermatitis in sensitized individuals, and chemical burns at 37%. Once a person is sensitized, very low concentrations can trigger dermatitis or asthma.
- Expected by-products: 37% formaldehyde contains 10-15% methanol as a stabilizer and evolves methanol vapor as well as formaldehyde. Cold or aged solutions can precipitate paraformaldehyde solids, which release formaldehyde gas. Formaldehyde reacts with hydrochloric acid or chloride salts under some conditions to form bis(chloromethyl) ether, a potent carcinogen - never mix formaldehyde with hydrochloric acid.
- GHS classification (from MyChem / SDS): Acute toxicity Category 3 (oral, dermal, inhalation); Carcinogenicity Category 1A; Germ cell mutagenicity Category 2; Skin corrosion/irritation Category 1B; Serious eye damage/eye irritation Category 1; Skin sensitization Category 1; Specific target organ toxicity - single and repeated exposure Category 1; Flammable liquids Category 3 (Combustible Liquid Class II).

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- ALL handling of 37% formaldehyde stock, all dilution, all filling and decanting of fixative, and all opening of specimen containers is performed inside the certified chemical fume hood in FTR 209, with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S; verify current certification and airflow before starting.
- Handling of tissue already fixed and transferred to 70% ethanol may be performed at the bench, provided the tissue has been rinsed and the storage fluid is ethanol rather than formalin.
- A disposable absorbent bench liner is used inside the hood for all formaldehyde work and discarded as hazardous waste.
- Formaldehyde is kept away from ignition sources; the 37% solution is a Class II combustible liquid.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly. Their location is confirmed before starting work, and the path to them is kept clear.
- A formaldehyde-specific spill kit containing a formaldehyde neutralizer/absorbent product is maintained in FTR 209 in addition to the general spill kit.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.

- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because formaldehyde is corrosive, a sensitizer, and toxic by skin contact and absorption, additional protective clothing is REQUIRED where splashes or skin contact are foreseeable: a chemically resistant apron and disposable chemically resistant sleeves for any dilution of 37% stock, any transfer of more than 100 mL, any bulk specimen jar work, and all spill cleanup.
- Contaminated clothing is removed immediately and is not taken home. Formaldehyde-saturated lab coats are bagged and discarded as hazardous waste rather than laundered.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: 8 mil or heavier nitrile gloves (e.g., Microflex 93-260, Halyard Purple Nitrile-Xtra) for incidental-contact handling of diluted formalin. For handling the 37% stock, for dilutions, and for any task with foreseeable contact, wear butyl rubber or laminate (North Silver Shield) gloves over an inner nitrile glove.
- Thin (3-4 mil) examination nitrile gloves are not adequate for 37% formaldehyde.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with formaldehyde. Gloves must be inspected prior to use, including a check for pinholes. Replace gloves immediately on any contact.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with formaldehyde. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all formaldehyde work. A face shield worn over goggles is REQUIRED for any dilution of the 37% stock, any transfer of more than 250 mL, and all spill cleanup, because formaldehyde causes irreversible eye damage.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all work is confined to the certified chemical fume hood as written.
- Formaldehyde is NEVER permitted to be used outside of the chemical fume hood under this SOP, with the single exception of handling securely capped, externally clean containers of fixed specimens during transport or storage.
- If a task is proposed that cannot be conducted in the fume hood, work must stop and the PI must contact EH&S for an exposure assessment under WAC 296-856 before the task proceeds. EH&S will determine whether personal exposure monitoring, a designated regulated area, and medical surveillance are required.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (0.5 ppm) or the PEL/STEL has been exceeded or when there is a possibility that it will be exceeded. Where air-purifying respirators are appropriate, use a full-face respirator with formaldehyde-specific cartridges as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

- Program enrollment through the EH&S Respiratory Protection Program includes medical evaluation, training and fit testing, and must be completed before any respirator is worn.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: the 37% stock is dispensed in volumes of no more than 500 mL at a time. Working fixative is prepared in volumes of no more than 4 L. Any operation requiring more than 1 L of 37% stock requires prior written PI approval and EH&S consultation.
- Never add formaldehyde to hydrochloric acid or to chloride-containing acidic solutions (bis(chloromethyl) ether formation). Keep formaldehyde separated from strong acids, strong bases, strong oxidizers (copper(II) nitrate, sulfuric acid), and from phenol and isocyanates.
- Specimen jars containing formalin must be filled, sealed and externally rinsed inside the fume hood before being moved to storage. Jar exteriors are wiped with water and dried so that no formaldehyde residue is carried to the bench or shelf.
- All containers of formaldehyde and formalin - including specimen jars and squeeze bottles - must carry a label with the full chemical name, concentration, the words 'CANCER HAZARD' and 'MAY CAUSE ALLERGIC REACTION', the date and the preparer's initials, in accordance with WAC 296-856 labeling requirements.
- Formaldehyde work is performed only when at least one other person is present in or immediately adjacent to FTR 209. Unattended operations involving open formaldehyde are prohibited; closed jars of specimens in fixative on a shelf are not considered an unattended operation.
- A 'FORMALDEHYDE IN USE - CANCER HAZARD - PARTICULARLY HAZARDOUS SUBSTANCE' sign is posted on the fume hood sash while work is in progress.
- Storage: the gallon of 37% formaldehyde is stored in its original container inside a chemically resistant secondary containment tray in the dedicated corrosives/toxics cabinet in FTR 209, below eye level, tightly closed, away from heat and light. Do not refrigerate the 37% stock - cold promotes paraformaldehyde precipitation. Specimen jars in formalin are stored in a labeled, closed cabinet with secondary containment; the storage cabinet carries a formaldehyde hazard sign.
- Transport: the gallon container is carried in a bottle carrier and is not transported outside FTR 209. Specimen jars are transported in a closed, leak-proof secondary container.
- Clean the chemical fume hood upon completion of tasks with water followed by a detergent solution and then 70% ethanol; wipe dry.
- Clean all contaminated surfaces with water and detergent solution and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with formaldehyde must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.

- Small spill - less than 100 mL of DILUTED formalin (10% or less) inside the fume hood: may be handled by trained lab staff wearing goggles, face shield, nitrile plus butyl gloves, lab coat with apron and sleeves. Cover with formaldehyde-neutralizing absorbent from the spill kit, work from the outside inward, allow the manufacturer-specified contact time, and collect the residue into a screw-top waste container inside the hood.
- Any spill of the 37% stock of any volume, any diluted spill greater than 100 mL, and ANY formaldehyde spill outside the fume hood: DO NOT clean up. Evacuate FTR 209 immediately, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S.
- Because formaldehyde is a Category 1A carcinogen with a low STEL, a spill outside a hood can exceed the 2 ppm STEL almost immediately in a closed room. Evacuation is the default response.
- Liquid spills are the expected form. If dried paraformaldehyde solids are found around a container neck or cap, do not brush or scrape them dry - dampen with water inside the fume hood and wipe up with absorbent, which prevents generating a formaldehyde-releasing dust.
- PPE required for cleanup: chemical splash goggles AND face shield, butyl or laminate gloves over 8 mil nitrile, buttoned lab coat with chemically resistant apron and disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - FORMALDEHYDE SPILL' on the door for any spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after neutralizer/absorbent removal, clean the spill area thoroughly with detergent solution followed by clean water, then wipe with 70% ethanol and dry.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, place in a screw-top container, label with a UW Hazardous Waste Label listing 'formaldehyde - spill debris', and submit a collection request to EH&S promptly.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification). Formaldehyde exposures above the Action Level trigger additional EH&S reporting and medical surveillance obligations under WAC 296-856.

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Inhalation exposure: move the person out of the contaminated area to fresh air immediately; get medical help. Any person with cough, wheeze or chest tightness after formaldehyde exposure must be evaluated the same day - delayed pulmonary edema is possible.
- Skin exposure: use the nearest safety shower for 15 minutes; stay under the shower and remove all contaminated clothing including shoes; use a clean lab coat or spare clothing for cover-up. 37% formaldehyde causes chemical burns; seek medical evaluation for any skin contact with the stock solution.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open, then go immediately to the Emergency Department. Formaldehyde causes irreversible eye damage; do not delay.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1 immediately.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (formaldehyde 37%, contains methanol), dose, route of exposure, and time since exposure. State that formaldehyde is a Category 1A carcinogen and a sensitizer. Bring the SDS and this SOP to the Emergency Department.

- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.
- Any employee who develops signs or symptoms of formaldehyde overexposure, or of sensitization (recurrent dermatitis, wheeze, or eye and airway irritation on re-exposure), is entitled to a medical examination under WAC 296-856. Report symptoms to the PI and EH&S.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused or expired 37% formaldehyde stock; (b) spent fixative (10% formalin) with and without tissue; (c) formaldehyde-contaminated aqueous rinses; (d) formaldehyde-contaminated solid debris (tips, wipes, liners, gloves, absorbent) and fixed tissue.
- Formaldehyde and formalin, including dilute spent fixative, are hazardous waste. Do NOT pour any formaldehyde-containing solution down the drain, in any concentration. In-lab neutralization or treatment of formaldehyde waste is NOT authorized under this SOP.
- Accumulate liquid waste at the point of generation, inside the fume hood, in a sturdy, chemically compatible HDPE or glass container with a securely closable screw-top lid, kept closed except when actively adding waste, and in secondary containment.
- Keep formaldehyde waste SEPARATE from acid waste (never mix with hydrochloric acid), from base waste, from oxidizer waste, and from halogenated solvent waste.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage (e.g., 'formaldehyde 4%, methanol 1%, water 95%, animal tissue'), starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Fixed tissue is drained inside the fume hood, the fixative collected as liquid hazardous waste, and the tissue either transferred to 70% ethanol for archiving or bagged, double-contained and submitted to EH&S as formaldehyde-contaminated solid waste. Never autoclave formaldehyde-contaminated material - it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Update the MyChem inventory when the formaldehyde container is emptied, received, or transferred. Consider surplusing unneeded formaldehyde rather than storing it indefinitely.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials.

Section 7 - Protocol

- Procedure A - Preparation of 10% neutral buffered formalin from 37% stock. Inside the certified fume hood, on an absorbent liner, wearing goggles, face shield, butyl over nitrile gloves, lab coat, apron and sleeves: measure 100 mL of 37% formaldehyde into a graduated cylinder, add it slowly to approximately 800 mL of phosphate buffer (or filtered seawater for marine tissue) in a 1 L bottle, bring to volume, cap and invert to mix. Label the bottle with 'FORMALIN

10% (3.7% formaldehyde) - CANCER HAZARD - MAY CAUSE ALLERGIC REACTION', the date and initials. Recap and return the 37% stock to the cabinet before proceeding.

- Procedure B - Fixation of bivalve or finfish tissue. Inside the fume hood, place dissected tissue into a labeled jar and add 10% formalin at approximately 10 volumes of fixative per volume of tissue. Cap the jar securely inside the hood, wipe the exterior with a damp wipe, dry, and move the jar to the formaldehyde storage cabinet. Fix for the interval specified by the protocol (typically 24-48 h at room temperature).
- Procedure C - Transfer of fixed tissue to 70% ethanol. Inside the fume hood, open the jar, decant the spent formalin into the labeled formaldehyde liquid waste container, rinse the tissue with phosphate buffer (collecting the rinse as formaldehyde waste), and transfer the tissue to a fresh labeled jar of 70% ethanol. Cap, wipe the exterior, and remove from the hood.
- Prohibited: mixing formaldehyde with hydrochloric acid or with bleach; heating formaldehyde; using formaldehyde outside the fume hood; using formaldehyde by untrained or unauthorized personnel; drain disposal of any formaldehyde-containing liquid.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs). Refer also to the UW EH&S formaldehyde use SOP and to WAC 296-856.
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - formaldehyde is used only on tissue harvested from euthanized animals, never administered to live animals. If formaldehyde is ever used on live animals or in a way that exposes live animals, the use will be documented and approved by IACUC and this SOP revised beforehand. Personnel handling animals prior to euthanasia are not exposed to formaldehyde under this SOP.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with formaldehyde must be trained on this SOP prior to starting work. They must also review the formaldehyde SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Formaldehyde is a particularly hazardous substance AND a substance-specific regulated chemical. Written authorization from the Principal Investigator is required before any individual may work with formaldehyde, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) formaldehyde-specific training meeting WAC 296-856 requirements, provided at initial assignment and at least annually thereafter, covering the contents of the standard, the SDS, the health hazards including carcinogenicity and sensitization, signs and symptoms of exposure, safe work practices, PPE, emergency procedures and the medical surveillance program; (4) documented review of this SOP.
- A worker must complete at least two supervised runs of Procedures A-C, observed and signed off by the PI or a designated experienced lab member, before performing formaldehyde work independently for the first time.

- EH&S must be contacted for an exposure assessment before any new or expanded formaldehyde operation begins. Medical surveillance is required for any employee exposed at or above the Action Level (0.5 ppm 8-hr TWA) or the STEL (2 ppm), and for any employee who develops signs or symptoms of overexposure; enrollment is arranged through EH&S and the Employee Health Center.
- A medical examination must be completed prior to any respirator use.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider and are encouraged to discuss alternative task assignment with the PI. Such consultation is voluntary and confidential.

Section 10 - Decontamination

- Fume hood: at the end of every formaldehyde session, remove and bag the absorbent bench liner as hazardous solid waste, then wipe the entire hood work surface and inside of the sash with water, then detergent solution, then 70% ethanol; allow to dry with the hood running.
- Equipment: wipe pipettors, graduated cylinders, funnels, forceps and dissection tools with detergent solution and water, then 70% ethanol. Dissection tools used on fixed tissue are washed with detergent and water inside the hood before removal.
- Glassware and specimen jars: rinse three times inside the fume hood, collecting all rinses in the formaldehyde liquid waste container, then wash with detergent and water and rinse three times with deionized water.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209. The formaldehyde designated area (Section 11) and the formaldehyde storage cabinet are wiped down as described above at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: water, laboratory detergent (Alconox or Contrad) solution, 70% ethanol, lint-free wipes, disposable absorbent bench liners, and a commercial formaldehyde neutralizer for spills. Do not use bleach on formaldehyde residues.

Section 11 - Designated Area

- DESIGNATED AREA: the certified chemical fume hood in FTR 209 is the sole designated area for all open handling, dilution, dispensing and decanting of formaldehyde and formalin.
- The designated area is posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - FORMALDEHYDE - CANCER HAZARD - MAY CAUSE ALLERGIC REACTION' sign listing the PI, contact number, required PPE and the authorization requirement, consistent with WAC 296-856 signage requirements.
- Formaldehyde waste containers are kept within the designated area, in secondary containment, inside the fume hood.
- Formaldehyde stock and sealed specimen jars are stored in the posted corrosives/toxics cabinet in FTR 209 with secondary containment; that cabinet is a designated storage location and carries the same hazard sign.
- No other bench, room or piece of equipment in the Roberts Lab space is authorized for open formaldehyde handling. Sealed, externally clean specimen jars may be carried through the lab in a secondary container.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.

- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

[illegible]